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	1	2	3	4	5	6	7	8
A	SHEET INDEX							
	PAGE		PAGE NAME		REVISION			
	==BIODIESEL_PLANT=H00&GAD/1		COVER SHEET		0			
	==BIODIESEL_PLANT=H00&GAD/2		INDEX OF SHEETS		0			
B	==BIODIESEL_PLANT=H00&GAD/3		GENERAL CHARACTERISTICS		0			
	==BIODIESEL_PLANT=H00&GAD/4		REFERENCE DESIGNATIONS		0			
	==BIODIESEL_PLANT=H00&GAD/5		REFERENCE DESIGNATIONS		0			
	==BIODIESEL_PLANT=H00&GAD/6		REFERENCE DESIGNATIONS		0			
	==BIODIESEL_PLANT=H00&GAD/7		REFERENCE DESIGNATIONS		0			
	==BIODIESEL_PLANT=H00&GAD/8		SINGLE LINE DIAGRAM		0			
	==BIODIESEL_PLANT=H00&GAD/9		SINGLE LINE DIAGRAM		0			
	==BIODIESEL_PLANT=H00&GAD/10		INSTALLATION RULES		0			
	==BIODIESEL_PLANT=H00&GAD/11		FRONT VIEW & GA		0			
C	==BIODIESEL_PLANT=H00&GAD/12		FOUNDATION FRAMES		0			
	==BIODIESEL_PLANT=H00&GAD/13		FOUNDATION FRAMES DETAILS		0			
	==BIODIESEL_PLANT=H00&GAD/14		FOUNDATION FRAMES DETAILS		0			
	==BIODIESEL_PLANT=H00&GAD/15		SECTION VIEW		0			
	==BIODIESEL_PLANT=H00&GAD/16		SECTION VIEW		0			
	==BIODIESEL_PLANT=H00&GAD/17		SECTION VIEW		0			
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				Date Prep.	26.03.2020	PROJECT TITLE		
				Prepared	YR	BIODIESEL AND FLEXTech PLANT		
0	Issued for Approval	26.03.2020	YR	Checked	AW			
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE320003	
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(IN COMPLIANCE WITH STANDARD IEC 62271-200)

NOTES

SUPPLIED IMPLEMENTS

IDENTIFICATION OF CONDUCTORS

Diagram illustrating a 3-to-2-to-3 decoder. The input is a 3-bit bus labeled $-XDA$ with bits 9, 10, and 11. The output is a 3-bit bus labeled $-XDA:10$ with bits 10, 11, and 12. The diagram shows a 2-bit intermediate bus labeled $-XDA:10$ with bits 10 and 11, which is connected to the output bus.

			Date Prep.	26.03.2020	PROJECT TITLE BIODIESEL AND FLEXTech PLANT	ABB	EPMV	GENERAL CHARACTERISTICS SLD & GA Drawing BIODIESEL PLANT	Scale		= H00	
			Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD	
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						PT. ABB Sakti Industri						

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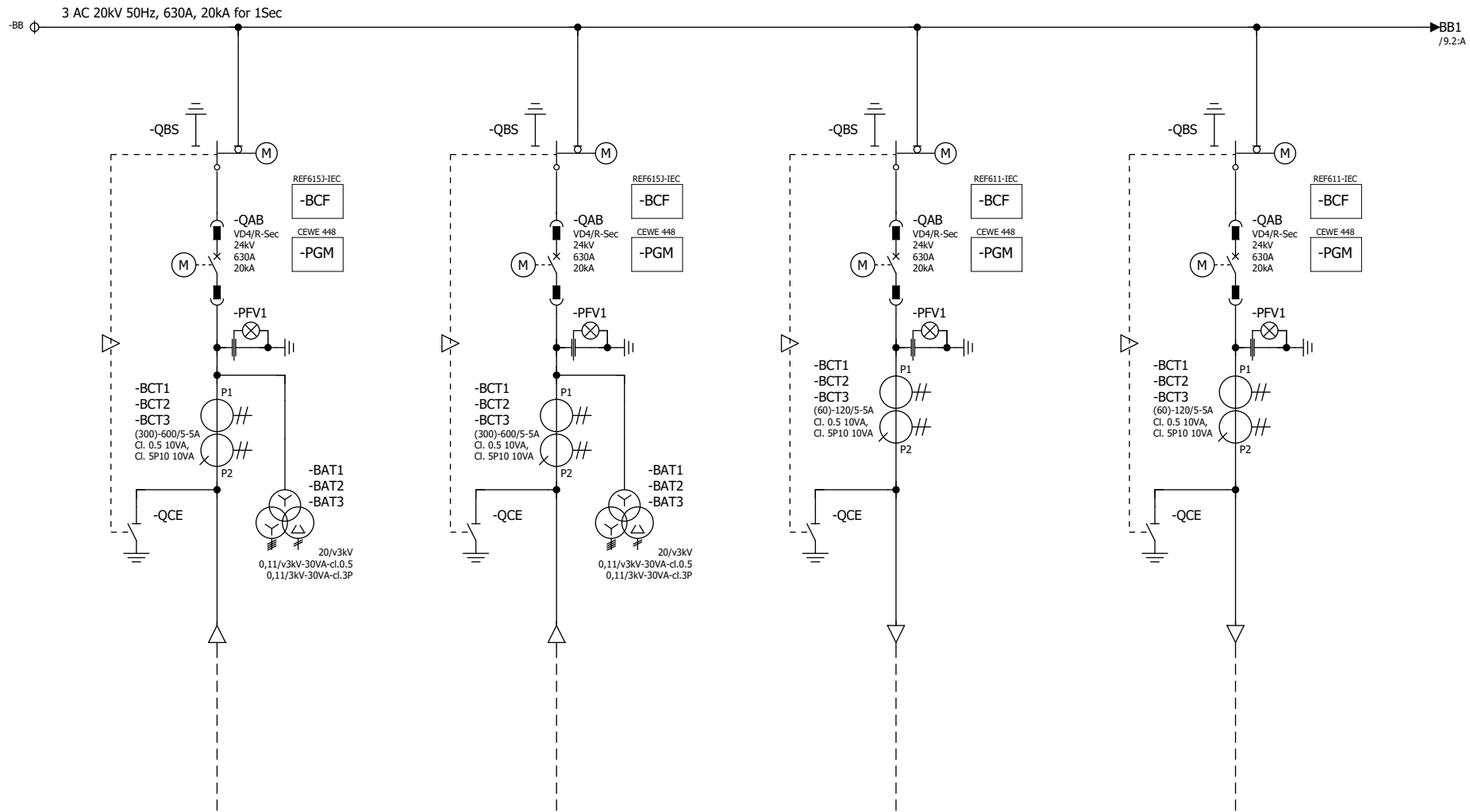
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A	-BG131	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION						-CA	CAPACITORS							A
	-BG132	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION						-CA1	CAPACITOR FOR FRONT VENTILATOR							
	-BG133	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)						-CA2	CAPACITOR FOR REAR VENTILATOR							
	-BG141...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION						-EA1	LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.							
	-BG142	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION						-EA2	LIGHTING LAMP LOCATED IN CABLE COMPARTMENT							
B	-BG143...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION						-EA3	LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE							B
	-BG151	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION						-EA4	LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT							
	-BG152	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION						-EB1	HEATER LOCATED IN CABLE COMPARTMENT							
	-BG153	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION						-EB2	HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT							
	-BGK	POSITION SWITCH OF KEY LOCK						-EB3	HEATER LOCATED IN MOTOR CABINET							
C	-BGL1	POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1						-EB4	HEATER FOR VENTILATION TEST							C
	-BGS1....-BGS4	POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS						-EB5	HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.							
	-BGS5	PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION						-EB7	HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE							
	-BGT1	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION						-EB8	HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE							
	-BGT2	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION						-FA1	SURGE ARRESTER LOCATED ON PHASE L1							
D	-BGT3	POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION						-FA2	SURGE ARRESTER LOCATED ON PHASE L2							D
	-BGT4	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION						-FA3	SURGE ARRESTER LOCATED ON PHASE L3							
	-BGT5	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION						-FCD	FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS							
	-BGT6	POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION						-FCF1	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1							
	-BGT7	POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION						-FCF2	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2							
E	-BGT8	POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION						-FCF3	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3							E
	-BGT11	PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION						-FCM1	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER							
	-BGT12	PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION						-FCM2...-FCM10	MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS							
	-BGV1	POSITION SWITCH OF FRONT VENTILATOR						-FCM11	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1							
	-BGV2	POSITION SWITCH OF REAR VENTILATOR						-FCM12	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2							
F	-BM	HYGROSTAT						-FCM13	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3							F
	-BPA4	PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-FCT	THERMAL OVERLOAD RELEASES							
	-BPA5	PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GA	GENERATORS							
	-BPA6	PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ1	FRONT VENTILATOR							
	-BPA7	PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ2	REAR VENTILATOR							
G	-BPS1	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER						-GQ3	VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK							G
	-BPS2	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2						-KFA1	AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE							
	-BPS3	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3						-KFA2	AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE							
	-BR	FLAME DETECTORS, SMOKE DETECTORS						-KFA3....-KFA9	AUXILIARY RELAYS OR CONTACTORS							
	-BT	THERMOSTAT						-KFC	CLOSING RELAYS OR CONTACTORS							
H	-BT1	THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS						-KFI	INTEGRATED CIRCUITS							H
	-BT2	THERMOSTAT FOR ENERGIZATION OF VENTILATORS						-KFL	LOCKOUT RELAY							
	-BT3	THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM						-KFM	MICROPROCESSORS							
	-BU	BUCHHOLZ RELAY						-KFN	ANTIPUMPING RELAYS							
	-BX1	UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING						-KFO	OPENING RELAYS OR CONTACTORS							
I	-BX2	ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING						-KFP	PROGRAMMABLE LOGIC CONTROLLERS (PLC)							I
	-BZ,-BZ4	MULTIFUNCTION SENSORS						-KFR	RECLOSING RELAYS							
	-BZ1	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1						-KFS	SYNCHRONIZING DEVICES							
	-BZ2	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2						-KFT	AUXILIARY TIME RELAYS, DELAY ELEMENTS							
	-BZ3	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3						-KFZ	CONTROLLERS							
J								-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER							J
								-MAD	MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD							
								-MAD1	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1							
								-MAD2	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2							
								-MAD9	MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS							
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A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE						-QBH	MANUAL CIRCUIT-BREAKERS							A
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2						-QBL	LINKS							
	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS						-QBM	MINIATURE SWITCH-DISCONNECTORS							
	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING						-QBS	SWITCH-DISCONNECTORS							
	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT						-QBT	TRUCK							
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER						-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING							
	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS						-QCE	EARTHING SWITCH							
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER						-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)							
	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER						-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)							
	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER						-QQ	CLUTCHES							
B	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS						-RAA	ANTIFERRORESONANCE RESISTOR							B
	-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER						-RAD	DIODES							
	-PFB	BLUE SIGNAL LAMPS						-RAI	INDUCTORS							
	-PFF	FLAG RELAYS						-RAR	RESISTORS							
	-PFG	GREEN SIGNAL LAMPS						-RF	FILTERS							
	-PFR	RED SIGNAL LAMPS						-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING							
	-PFS	SHORT CIRCUIT INDICATORS						-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT							
	-PFV	VOLTAGE INDICATORS						-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE							
	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE						-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION							
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE						-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD							
C	-PFW	WHITE SIGNAL LAMPS						-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1							C
	-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS						-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2							
	-PFY	YELLOW SIGNAL LAMPS						-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2							
	-PGA	AMMETERS						-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS							
	-PGC	COUNTERS						-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4							
	-PGF	FREQUENCYMETERS						-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5							
	-PGH	HOURMETERS						-SFA	AMMETRIC SWITCHES							
	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE						-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS							
	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)						-SFL	LOCKING CONTACTS							
	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)						-SFO	OPENING PUSH-BUTTONS							
D	-PGJ	ACTIVE ENERGY METERS						-SFR	RESET PUSH-BUTTONS							D
	-PGK	REACTIVE ENERGY METERS						-SFS	SELECTOR SWITCHES							
	-PGM	MULTIFUNCTION INDICATORS						-SFT	TEST PUSH-BUTTONS							
	-PGP	POWER-FACTOR METERS						-SFU	UNLOCKING PUSH-BUTTONS							
	-PGQ	VARMETERS						-SPV	VOLTMETRIC SWITCHES							
	-PGS	SYNCHRONOSCOPES						-TA	POWER TRANSFORMERS							
	-PGV	VOLTMETERS						-TB	CONVERTER							
	-PGW	WATTMETERS						-TB1	RECTIFIER FOR FIRST OPENING RELEASE							
	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)						-TB2	RECTIFIER FOR SECOND OPENING RELEASE							
	-QAB	CIRCUIT-BREAKERS						-TB3	RECTIFIER FOR CLOSING RELEASE							
E	-QAC	CONTACTORS (FOR POWER)						-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING							E
	-QBD	DISCONNECTORS						-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT							
	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE							
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER							
	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)														
	-QBD4	TWO-WAY DISCONNECTOR														
	-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER														
F					Date Prep.	26.03.2020	PROJECT TITLE		ABB	EPMV	REFERENCE DESIGNATIONS SLD & GA Drawing BIODIESEL PLANT		Scale		= H00	F
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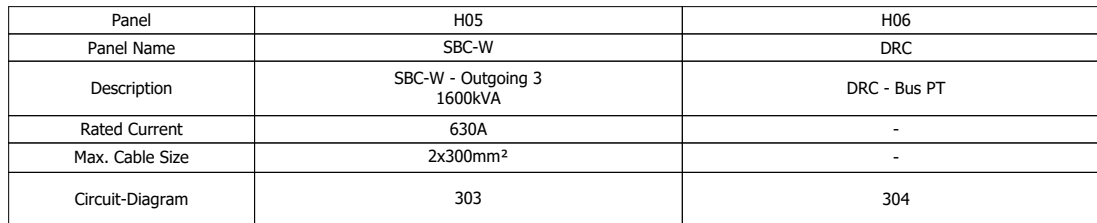
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A	-TFA	ACTIVE POWER TRANSDUCERS						-XD11	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS							A									
	-TFC	CURRENT TRANSDUCERS						-XD12	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING																
	-TFF	FREQUENCY TRANSDUCERS						-XD13	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION																
	-TFJ	ACTIVE ENERGY TRANSDUCERS						-XD14	TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS																
	-TFK	REACTIVE ENERGY TRANSDUCERS						-XD16	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS																
	-TFM	MULTIFUNCTION TRANSDUCERS						-XD17	TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS																
	-TFP	POWER-FACTOR TRANSDUCERS						-XD18	TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS																
	-TFQ	REACTIVE POWER TRANSDUCERS						-XD19	TERMINAL BLOCK FOR SPECIAL USE																
	-TFS	SIGNAL CONVERTERS						-XDJ	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD																
	-TFV	VOLTAGE TRANSDUCERS						-XDJ1	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS																
B	-WA	BUSBARS						-XDM	SEALABLE TERMINAL BLOCK FOR MEASUREMENT							B									
	-WBC	POWER CABLES						-XDS	CURRENT SOCKETS																
	-WE	EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.						-XDT	TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB																
	-WGC	CONTROL CABLES						-XDU	CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK																
	-WGS	COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)						-XDU1	TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK																
	-WH	OPTICAL FIBRES						-XDV	TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS																
	-XDA	TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS						-XDV1	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1																
	-XDA1	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1						-XDV2	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2																
	-XDA2	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2						-XDV3	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3																
	-XDA3	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3						-XDV4	CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR																
C	-XDA5	CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER						-XDX	SUPPORT TERMINAL BLOCK							C									
	-XDB	CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS						-XE1	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE																
	-XDB1	CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)						-XE2	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE																
	-XDB2	CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER						-XE5	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE																
	-XDB10...-XDB89	CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS						-XE6	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE																
	-XDB7	CONNECTOR FOR TRUCK POSITION CONTACTS						-XE7	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE																
	-XDB91	CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR						-XE8	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE																
	-XDB92	CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR						-XE9	EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT																
	-XDB93	CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR																							
	-XDB95	CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET																							
D	-XDC	CUSTOMER TERMINAL BLOCK														D									
	-XDC1	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER																							
	-XDC2	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR																							
	-XDC3	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT																							
	-XDC4	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS																							
	-XDC5	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER																							
	-XDE	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE																							
	-XDE1	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE																							
	-XDE2	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2																							
	-XDE3	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2																							
E	-XDF	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS														E									
	-XDF1	CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR																							
	-XDF2	CONNECTOR FOR REAR VENTILATOR CIRCUITS																							
	-XDF3	CONNECTOR FOR FRONT VENTILATOR CIRCUITS																							
	-XDH	TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)																							
	-XDI	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)																							
	F					Date Prep.	26.03.2020	PROJECT TITLE		ABB	EPMV	REFERENCE DESIGNATIONS SLD & GA Drawing BIODIESEL PLANT		Scale			= H00	F							
					Prepared	YR	BIODIESEL AND FLEXTech PLANT							Customer	PT Multimas Nabati Asahan	& GAD									
0		Issued for Approval	26.03.2020	YR	Checked	AW								PT. ABB Sakti Industri						Drawing Nbr.		Sh. 7			
Ind.		Revision	Date	Name	Approved	AI														Based on: BE320003					
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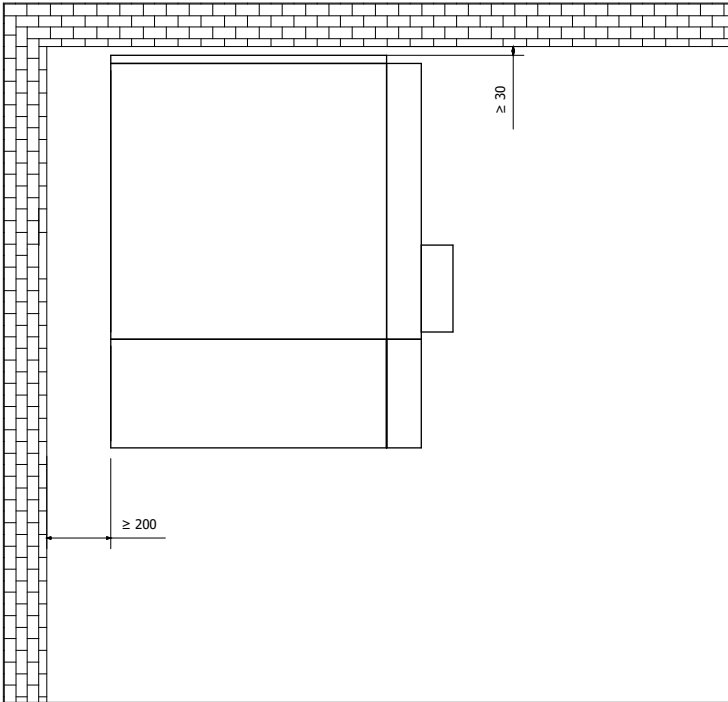
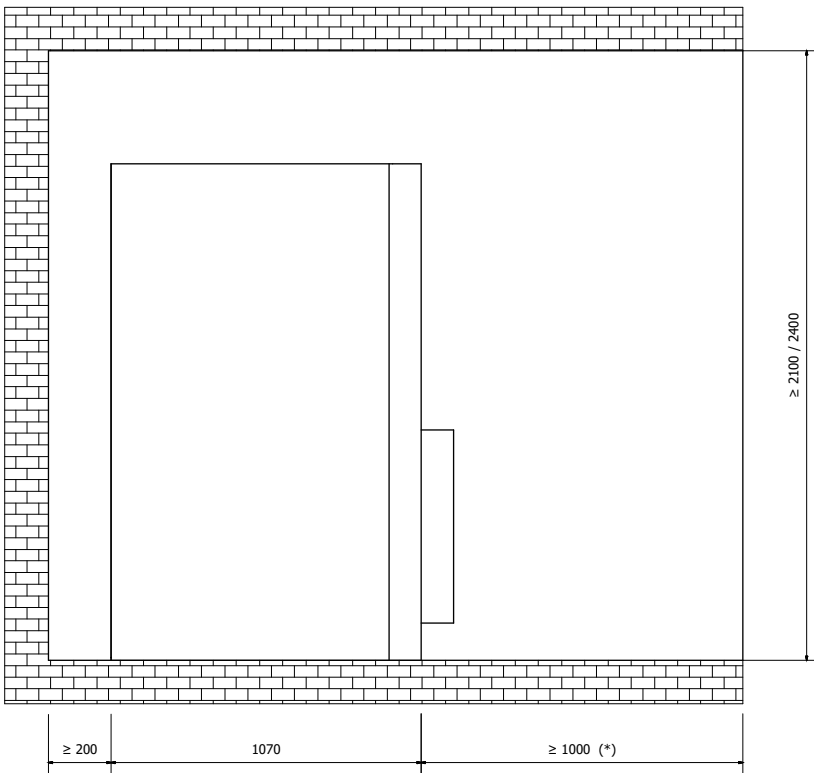
Panel	H01	H02	H03	H04
Panel Name	SBC-W	SBC-W	SBC-W	SBC-W
Description	SBC-W - Incoming 1	SBC-W - Incoming 2	SBC-W - Outgoing 1 1600kVA	SBC-W - Outgoing 2 1600kVA
Rated Current	630A	630A	630A	630A
Max. Cable Size	2x300mm ²	2x300mm ²	2x300mm ²	2x300mm ²
Circuit-Diagram	301	302	303	303

Ind.	Revision	Date	Name	Approved	AI	Based on:	BE320003	PROJECT TITLE	BIODIESEL AND FLEXTECH PLANT	ABB EPMV	PT. ABB Sakti Industri	SINGLE LINE DIAGRAM SLD & GA Drawing BIODIESEL PLANT	Scale	= H00	Customer	PT Multimas Nabati Asahan	& GAD	Drawing Nbr.	Sh. 8	Customer Doc. Nbr.	Cont. 17
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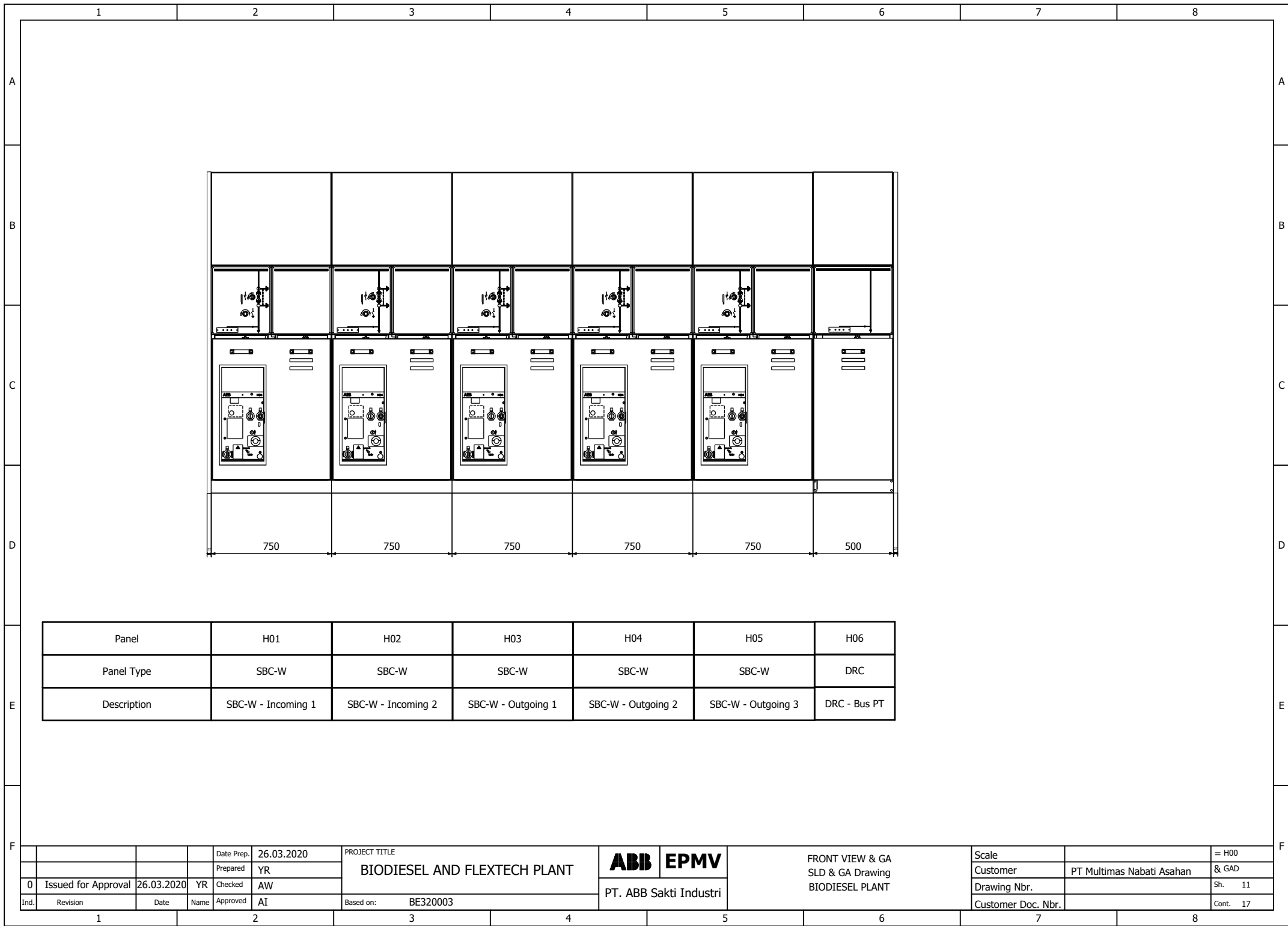


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			Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	Issued for Approval	26.03.2020	YR	Checked AW					Drawing Nbr.		Sh. 9
Ind.	Revision	Date	Name	Approved AI	Based on: BE320003	PT. ABB Sakti Industri		Customer Doc. Nbr.		Cont. 17	

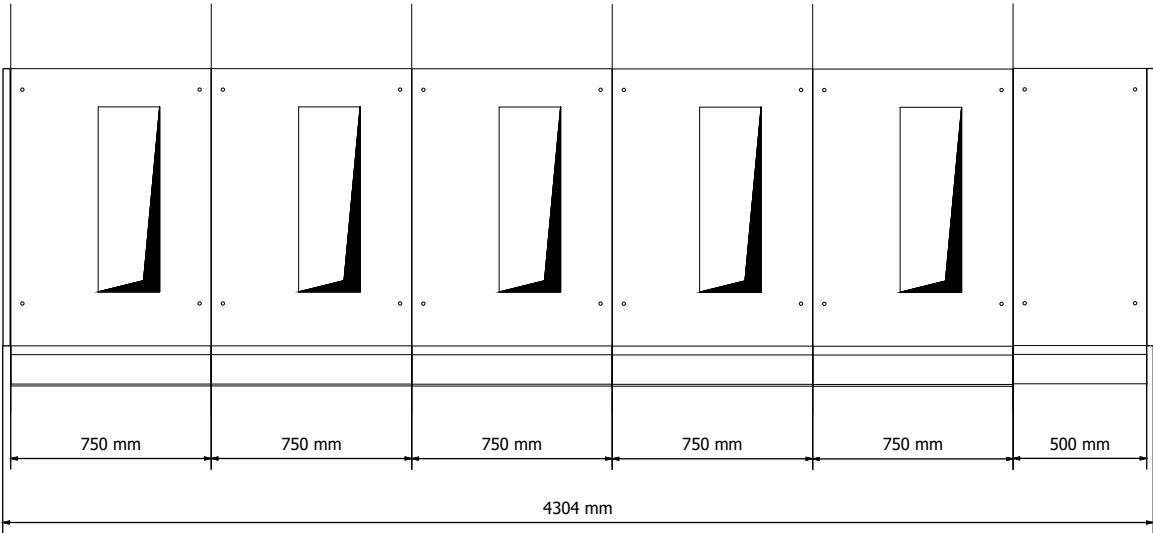
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RULES FOR INSTALLATION IN THE ROOM																																																																																	
MINIMUM DISTANCES TO BE RESPECTED DURING INSTALLATION																																																																																	
  (*) ≥ 1300 mm at least for units with circuit-breaker																																																																																	
- IAC A-F SOLUTION UP TO 16kA 1sec - ATTENTION : NO ACCESS TO REAR AND LATERAL OF THE SWG ROOM WHILE SWG IS IN SERVICE																																																																																	
<table><tr><td colspan="2"></td><td colspan="2"></td><td colspan="2">Date Prep. 26.03.2020</td><td colspan="2">PROJECT TITLE</td><td colspan="2">ABB EPMV</td><td colspan="2">INSTALLATION RULES</td><td colspan="2">Scale</td><td colspan="2">= H00</td></tr><tr><td colspan="2"></td><td colspan="2"></td><td colspan="2">Prepared YR</td><td colspan="2">BIO DIESEL AND FLEXTech PLANT</td><td colspan="2" rowspan="3">PT. ABB Sakti Industri</td><td colspan="2">SLD & GA Drawing</td><td colspan="2">Customer</td><td colspan="2">PT Multimas Nabati Asahan</td><td colspan="2">& GAD</td></tr><tr><td colspan="2">0</td><td colspan="2">Issued for Approval</td><td colspan="2">26.03.2020</td><td colspan="2">YR</td><td colspan="2">Checked AW</td><td colspan="2">Drawing Nbr.</td><td colspan="2"></td><td colspan="2">Sh. 10</td></tr><tr><td colspan="2">Ind.</td><td colspan="2">Revision</td><td colspan="2">Date</td><td colspan="2">Name</td><td colspan="2">Approved AI</td><td colspan="2">Based on: BE320003</td><td colspan="2">Customer Doc. Nbr.</td><td colspan="2">Cont. 17</td></tr></table>																				Date Prep. 26.03.2020		PROJECT TITLE		ABB EPMV		INSTALLATION RULES		Scale		= H00						Prepared YR		BIO DIESEL AND FLEXTech PLANT		PT. ABB Sakti Industri		SLD & GA Drawing		Customer		PT Multimas Nabati Asahan		& GAD		0		Issued for Approval		26.03.2020		YR		Checked AW		Drawing Nbr.				Sh. 10		Ind.		Revision		Date		Name		Approved AI		Based on: BE320003		Customer Doc. Nbr.		Cont. 17	
				Date Prep. 26.03.2020		PROJECT TITLE		ABB EPMV		INSTALLATION RULES		Scale		= H00																																																																			
				Prepared YR		BIO DIESEL AND FLEXTech PLANT		PT. ABB Sakti Industri		SLD & GA Drawing		Customer		PT Multimas Nabati Asahan		& GAD																																																																	
0		Issued for Approval		26.03.2020		YR				Checked AW		Drawing Nbr.				Sh. 10																																																																	
Ind.		Revision		Date		Name				Approved AI		Based on: BE320003		Customer Doc. Nbr.		Cont. 17																																																																	
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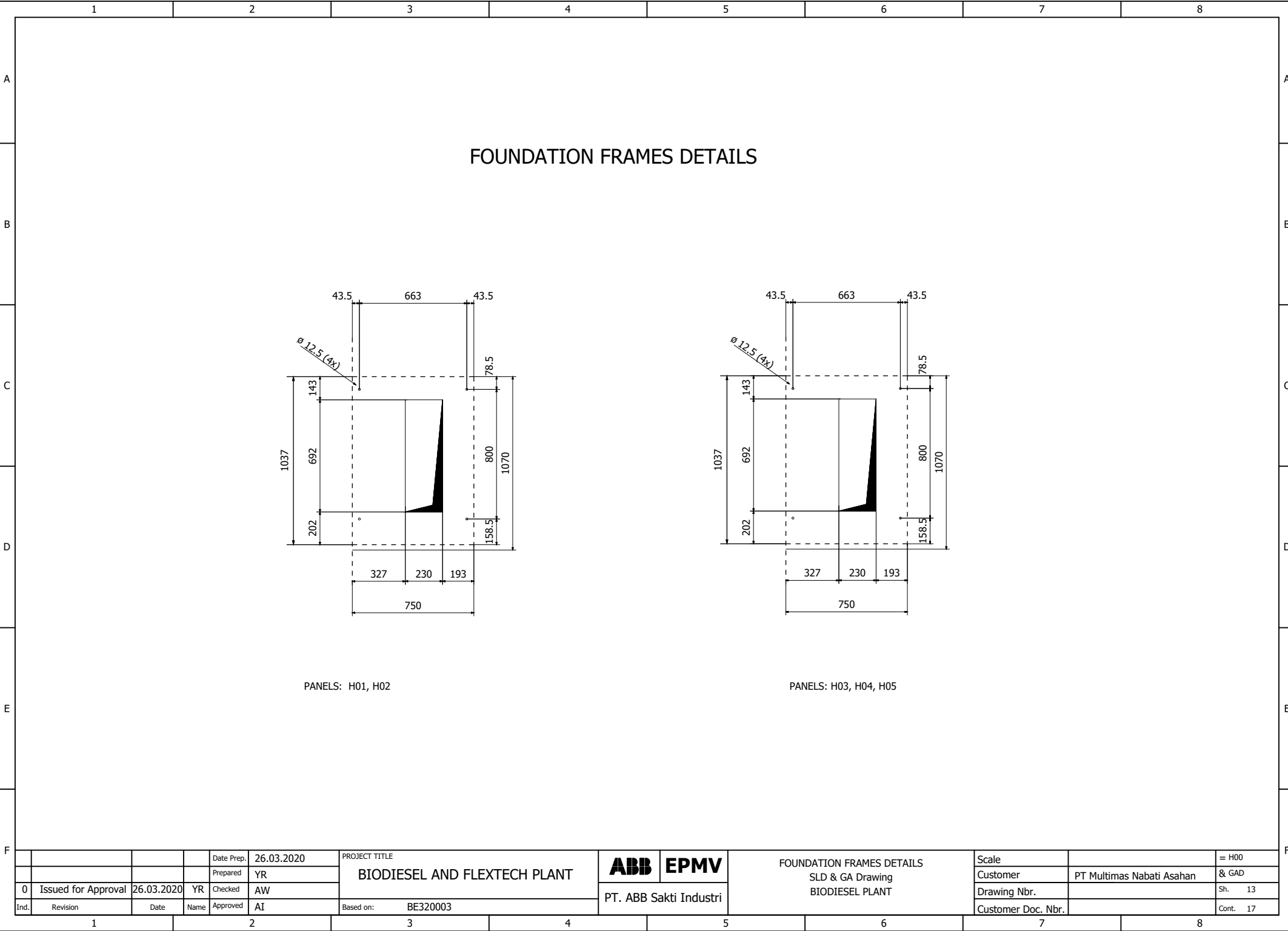
FOUNDATION FRAMES



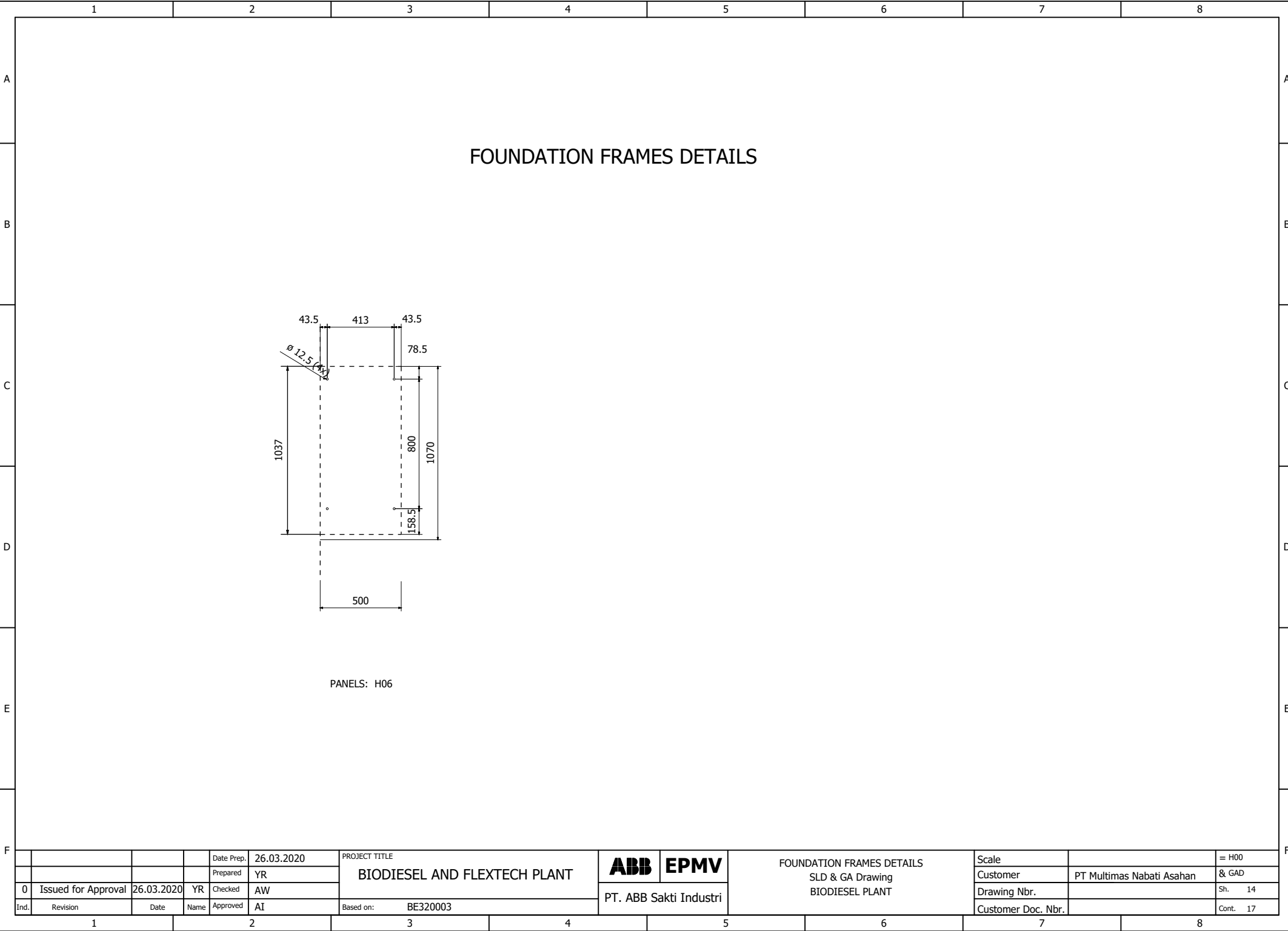
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Panel Type	SBC-W	SBC-W	SBC-W	SBC-W	SBC-W	DRC
Description	SBC-W - Incoming 1	SBC-W - Incoming 2	SBC-W - Outgoing 1	SBC-W - Outgoing 2	SBC-W - Outgoing 3	DRC - Bus PT

				Date Prep.	26.03.2020	PROJECT TITLE BIODIESEL AND FLEXTech PLANT	 	FOUNDATION FRAMES SLD & GA Drawing BIODIESEL PLANT	Scale		= H00
			Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	Issued for Approval	26.03.2020	YR	Checked	AW				Drawing Nbr.		Sh. 12
Ind.	Revision		Date	Name	Approved	AI	Based on: BE320003		Customer Doc. Nbr.		Cont. 17
							PT. ABB Sakti Industri				

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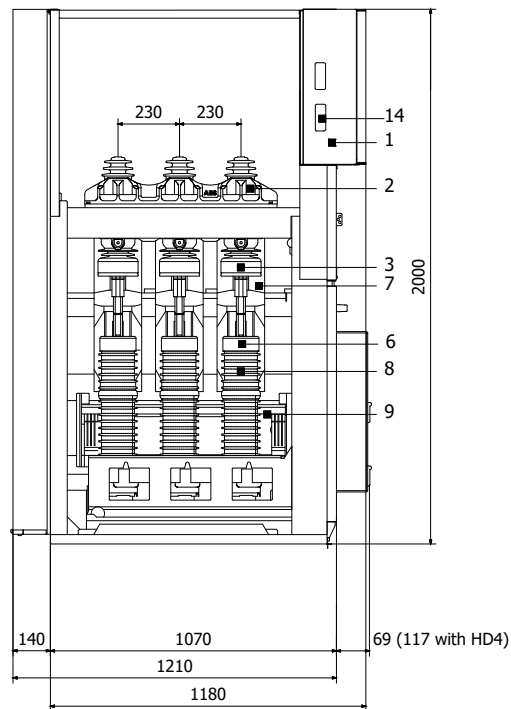


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SECTION VIEW



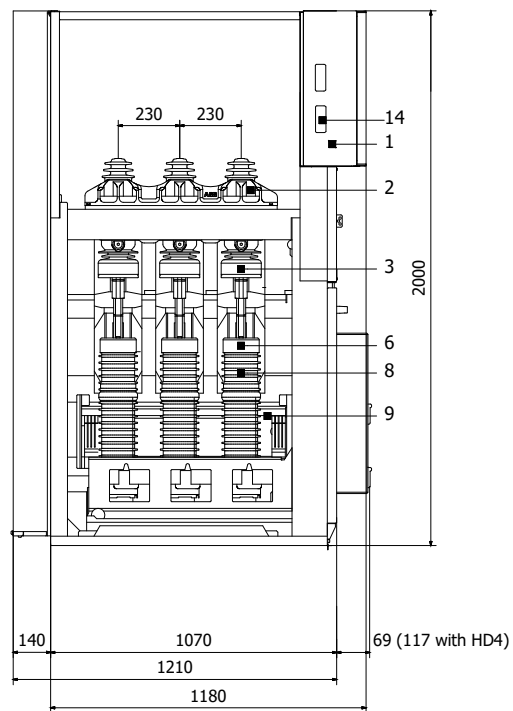
-	DISTANCE BETWEEN THE CABLE CONNECTION POINT AND THE FLOOR = 480mm
1	LVC
2	SWITCH DISCONNECTOR
3	LOWER COPPER BARS
6	CURRENT TRANSFORMER
7	VOLTAGE TRANSFORMER
8	CIRCUIT BREAKER
9	EARTHING-SWITCH
14	INTERPANEL WIRING PASSAGE

Panel : H01, H02

			Date Prep.	26.03.2020	PROJECT TITLE BIODIESEL AND FLEXTech PLANT	ABB	EPMV	SECTION VIEW SLD & GA Drawing BIODIESEL PLANT	Scale		= H00	
			Prepared	YR							Customer	PT Multimas Nabati Asahan
0	Issued for Approval	26.03.2020	YR	Checked		AW				Drawing Nbr.		Sh. 15
Ind.	Revision	Date	Name	Approved	AI	Based on: BE320003			PT. ABB Sakti Industri	Customer Doc. Nbr.		Cont. 17

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SECTION VIEW



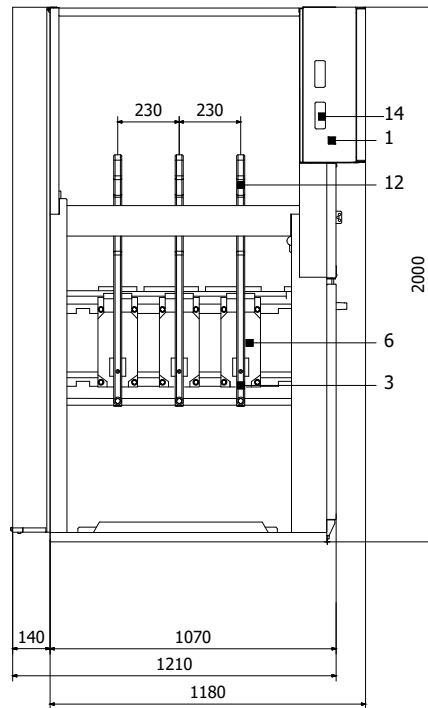
-	DISTANCE BETWEEN THE CABLE CONNECTION POINT AND THE FLOOR = 480mm
1	LVC
2	SWITCH DISCONNECTOR
3	LOWER COPPER BARS
6	CURRENT TRANSFORMER
8	CIRCUIT BREAKER
9	EARTHING-SWITCH
14	INTERPANEL WIRING PASSAGE

Panel : H03, H04, H05

			Date Prep.	26.03.2020	PROJECT TITLE BIODIESEL AND FLEXTech PLANT	ABB	EPMV	SECTION VIEW SLD & GA Drawing BIODIESEL PLANT	Scale		= H00	
			Prepared	YR							Customer	PT Multimas Nabati Asahan
0	Issued for Approval	26.03.2020	YR	Checked		AW				Drawing Nbr.		Sh. 16
Ind.	Revision	Date	Name	Approved	AI	Based on: BE320003			PT. ABB Sakti Industri	Customer Doc. Nbr.		Cont. 17

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SECTION VIEW



-	DISTANCE BETWEEN THE CABLE CONNECTION POINT AND THE FLOOR = 670mm
1	LVC
3	LOWER COPPER BARS
6	VOLTAGE TRANSFORMER
12	TOP COPPER BARS
14	INTERPANEL WIRING PASSAGE

Panel : H06

				Date Prep.	26.03.2020	PROJECT TITLE BIODIESEL AND FLEXTech PLANT	ABB	EPMV	SECTION VIEW SLD & GA Drawing BIODIESEL PLANT	Scale		= H00
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	Issued for Approval	26.03.2020	YR	Checked	AW		PT. ABB Sakti Industri	Drawing Nbr.			Sh.	17
Ind.	Revision	Date	Name	Approved	AI			Based on:		BE320003	Customer Doc. Nbr.	